

Technical Assessment (Head of Data Engineering)

Purpose

This assessment is designed to evaluate both your technical depth and strategic thinking as a Head of Data Engineering.

You will demonstrate your ability to design and build a modern data platform that supports analytics, machine learning, and decision-making in the agriculture sector.

Objectives

Through this exercise, you will demonstrate your ability to:

1. Design a scalable and governed data platform architecture.
2. Build a small, working data pipeline that showcases your technical skills.
3. Apply data management and quality practices relevant to agricultural data.
4. Communicate your design clearly through a short presentation and demo.

Scenario

You have joined an agriculture technology organisation that wants to become more data-driven.

The business currently collects data from multiple farms, sensors, and spreadsheets, but insights are inconsistent and difficult to scale.

You have been asked to design and demonstrate part of a modern data platform that can manage, process, and analyse agricultural data to support real-time insights and reporting.

What You Need to Do

Part 1 – Platform Design and Strategy

Objective: Show your high-level architectural and strategic thinking.

Task:

Design a modern data platform for the organisation described above. Your design should:

- Outline key design principles (scalability, security, interoperability, governance).

- Identify the core layers (ingestion, storage, processing, analytics, metadata).
- Recommend relevant technologies or patterns (cloud, orchestration, streaming, or data lakehouse).
- Explain how the platform would handle data lineage, metadata, and quality control.

Deliverable:

- A short presentation (5–7 slides in PDF or PowerPoint) describing your architecture, design rationale, and delivery approach.

Part 2 – Hands-On Data Engineering Task

Objective: Demonstrate your ability to build and document a working data pipeline.

Datasets provided:

1. **farms.csv**

Contains farm-level information.

farm_id	region	crop_type	area_hectares	owner_name	start_date
FARM_001	Angus	Barley	120	Smith Family Farms	2020-02-01

2. **sensor_readings.csv**

Contains IoT sensor readings from farms.

reading_id	farm_id	sensor_type	reading_value	reading_ts
READ_0001	FARM_001	SoilMoisture	32.5	2023-06-12 10:15:00

Task Instructions

1. Ingest both datasets into your environment (Python, PySpark, SQL, or dbt).
2. Clean and prepare the data:
 - Handle duplicates and missing values.
 - Ensure timestamps are standardised.
3. Transform the data to create:
 - **Average daily soil moisture and temperature per farm.**
 - **A list of farms showing low soil moisture (<30%) for three or more consecutive days.**
4. Save the output data in a structured format (Parquet, JSON, or CSV).
5. (Optional Bonus) Add:
 - A small chart or visualisation of moisture trends.
 - A metadata or lineage example showing data source, timestamp, and schema version.

Deliverables:

- Working code (Python notebook, scripts, or dbt project).
- Output data files.
- A README file describing how to run the code, key assumptions, and tools used.
- Optional visualisation or lineage sample.

Part 3 – Presentation & Demo

Objective: Present and explain your work during the interview.

Interview Format (90 minutes total):

- 30–40 minutes: Your presentation and live demo of the pipeline on your laptop.
- 50–60 minutes: Competency-based and follow-up questions.

Deliverables to Submit (3 days before the interview):

1. Presentation slides (5–7 slides).
2. Code and data outputs (zipped folder or GitHub link).
3. README file.

Timeline

- You will have **one week** to complete the exercise.
- Expected effort: **4–5 hours total**.
- Submit all materials **3 days before your scheduled interview**.

Evaluation Criteria

Area	Description	Weight
Architecture & Design	Clear, scalable, domain-relevant data platform design	25%
Technical Implementation	Working, well-structured data pipeline	25%
Data Quality & Lineage	Demonstrated validation and metadata awareness	20%
Analytical Insight	Ability to extract meaningful agricultural insights	15%
Communication & Documentation	Clarity, structure, and completeness of delivery	15%

Submission Format

Please submit the following **as a single email or shared link**:

1. Presentation (PDF or PowerPoint)
2. GitHub link containing your code and outputs
3. README / setup instructions
4. Any additional diagrams or notes (optional)